

Empirical Analysis of Heuristic Functions and Lookahead Depth in Adversarial Mancala AI

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Abstract

This report presents an exhaustive empirical analysis of Alpha-Beta Pruning decision engines applied to the deterministic zero-sum board game Mancala. Using custom heuristic weight configurations and move-ordering optimizations, six distinct evaluation functions (H_1 through H_6) were benchmarked across search depths 4, 6, and 8. A round-robin tournament structure comprising 100 matches per head-to-head pairing (4,500 total games simulated) was executed to measure win ratios, draw rates, and strategic stability. The data reveals critical trade-offs between static evaluation heuristic complexity and search depth scaling.

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1 Introduction

Mancala is a classic two-player turn-based strategy game characterized by perfect information and zero-sum outcomes. The primary objective is to collect more stones in one's store (Kalah) than the opponent by the time all playing bins are depleted.

To compute optimal moves, game engines rely on the **Minimax algorithm** augmented by **Alpha-Beta Pruning**. While exhaustive depth searching guarantees optimal play theoretically, time and memory limits necessitate truncating the search tree at a maximum depth d and evaluating non-terminal state quality using a static evaluation function $h(s)$.

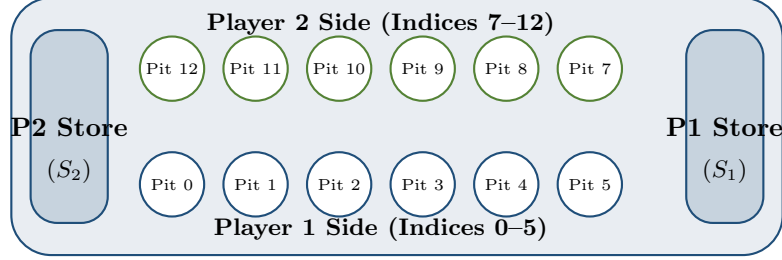


Figure 1: Schematic representation of the 14-bin Mancala board layout and indexing system.

2 System Architecture & Heuristic Definitions

2.1 Search Engine & Alpha-Beta Pruning

The decision engine uses Minimax search with dynamic move ordering and α - β pruning. Dynamic ordering sorts moves by evaluating immediate extra-turn opportunities and capture yields to maximize early pruning of suboptimal branches.

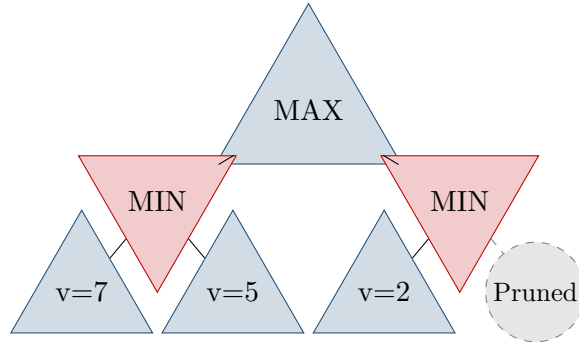


Figure 2: Alpha-Beta pruning branch evaluation illustrating search space reduction.

2.2 Mathematical Formulation of Evaluators

Six heuristic functions were constructed using specific feature combinations[cite: 1]:

1. **Heuristic 1 (H_1):** Storage Difference[cite: 1]

$$H_1(s) = S_1 - S_2 \quad (1)$$

2. **Heuristic 2 (H_2):** Storage + Side Difference[cite: 1]

$$H_2(s) = 1.0 \cdot (S_1 - S_2) + 0.5 \cdot (\text{Side}_1 - \text{Side}_2) \quad (2)$$

3. **Heuristic 3 (H_3):** Storage + Side + Extra Turn[cite: 1]

$$H_3(s) = 1.0 \cdot (S_1 - S_2) + 0.3 \cdot (\text{Side}_1 - \text{Side}_2) + 2.0 \cdot \mathbb{I}(\text{ExtraTurn}) \quad (3)$$

4. **Heuristic 4** (H_4): Storage + Side + Extra Turn + Captures[cite: 1]

$$H_4(s) = 1.0 \cdot (S_1 - S_2) + 0.3 \cdot (\text{Side}_1 - \text{Side}_2) + 1.5 \cdot \mathbb{I}(\text{ExtraTurn}) + 1.0 \cdot \text{Captured} \quad (4)$$

5. **Heuristic 5** (H_5): Storage + Close Stones + Extra Turn[cite: 1]

$$H_5(s) = 1.0 \cdot (S_1 - S_2) + 0.2 \cdot \text{StonesCloseToStorage} + 3.0 \cdot \mathbb{I}(\text{ExtraTurn}) \quad (5)$$

6. **Heuristic 6** (H_6): Win Proximity Model[cite: 1]

$$H_6(s) = 2.0 \cdot \left(\frac{1}{24 - S_1} - \frac{1}{24 - S_2} \right) + 0.3 \cdot (\text{Side}_1 - \text{Side}_2) + 1.5 \cdot \text{Captured} \quad (6)$$

3 Experimental Methodology

A round-robin tournament was executed across three search depths ($d \in \{4, 6, 8\}$)[cite: 2]. For each depth, all 15 unique pairs played 100 matches, alternating starting roles to eliminate first-mover advantage[cite: 1, 2].

4 Complete Empirical Results

4.1 Head-to-Head Breakdown: Depth 4

Table 1 outlines all 15 match outcomes at search depth 4[cite: 2].

Table 1: Round-Robin Results for Search Depth 4 (100 Games per Matchup)

Matchup (H-A vs H-B)	H-A Wins	H-B Wins	Draws	H-A Win %	H-B Win %
heuristic-1 vs heuristic-2	74	22	4	74%	22%
heuristic-1 vs heuristic-3	49	46	5	49%	46%
heuristic-1 vs heuristic-4	53	40	7	53%	40%
heuristic-1 vs heuristic-5	44	42	14	44%	42%
heuristic-1 vs heuristic-6	54	45	1	54%	45%
heuristic-2 vs heuristic-3	42	58	0	42%	58%
heuristic-2 vs heuristic-4	52	46	2	52%	46%
heuristic-2 vs heuristic-5	21	79	0	21%	79%
heuristic-2 vs heuristic-6	68	31	1	68%	31%
heuristic-3 vs heuristic-4	33	15	52	33%	15%
heuristic-3 vs heuristic-5	0	100	0	0%	100%
heuristic-3 vs heuristic-6	95	5	0	95%	5%
heuristic-4 vs heuristic-5	0	95	5	0%	95%
heuristic-4 vs heuristic-6	75	24	1	75%	24%
heuristic-5 vs heuristic-6	100	0	0	100%	0%

4.2 Head-to-Head Breakdown: Depth 6

Table 2 documents performance at search depth 6[cite: 2].

Table 2: Round-Robin Results for Search Depth 6 (100 Games per Matchup)

Matchup (H-A vs H-B)	H-A Wins	H-B Wins	Draws	H-A Win %	H-B Win %
heuristic-1 vs heuristic-2	91	5	4	91%	5%
heuristic-1 vs heuristic-3	56	33	11	56%	33%
heuristic-1 vs heuristic-4	49	49	2	49%	49%
heuristic-1 vs heuristic-5	53	42	5	53%	42%
heuristic-1 vs heuristic-6	76	22	2	76%	22%
heuristic-2 vs heuristic-3	0	100	0	0%	100%
heuristic-2 vs heuristic-4	18	69	13	18%	69%
heuristic-2 vs heuristic-5	0	100	0	0%	100%
heuristic-2 vs heuristic-6	99	1	0	99%	1%
heuristic-3 vs heuristic-4	76	24	0	76%	24%
heuristic-3 vs heuristic-5	36	62	2	36%	62%
heuristic-3 vs heuristic-6	100	0	0	100%	0%
heuristic-4 vs heuristic-5	37	62	1	37%	62%
heuristic-4 vs heuristic-6	50	50	0	50%	50%
heuristic-5 vs heuristic-6	50	48	2	50%	48%

4.3 Head-to-Head Breakdown: Depth 8

Table 3 provides metrics at lookahead depth 8[cite: 2].

Table 3: Round-Robin Results for Search Depth 8 (100 Games per Matchup)

Matchup (H-A vs H-B)	H-A Wins	H-B Wins	Draws	H-A Win %	H-B Win %
heuristic-1 vs heuristic-2	59	36	5	59%	36%
heuristic-1 vs heuristic-3	79	17	4	79%	17%
heuristic-1 vs heuristic-4	79	19	2	79%	19%
heuristic-1 vs heuristic-5	51	49	0	51%	49%
heuristic-1 vs heuristic-6	74	26	0	74%	26%
heuristic-2 vs heuristic-3	27	69	4	27%	69%
heuristic-2 vs heuristic-4	53	46	1	53%	46%
heuristic-2 vs heuristic-5	50	50	0	50%	50%
heuristic-2 vs heuristic-6	76	0	24	76%	0%
heuristic-3 vs heuristic-4	42	58	0	42%	58%
heuristic-3 vs heuristic-5	22	70	8	22%	70%
heuristic-3 vs heuristic-6	78	2	20	78%	2%
heuristic-4 vs heuristic-5	18	69	13	18%	69%
heuristic-4 vs heuristic-6	81	2	17	81%	2%
heuristic-5 vs heuristic-6	95	0	5	95%	0%

5 Graphical Performance & Comparative Analysis

5.1 Overall Win Comparison Across Depths

Figure 3 compares the aggregate wins for each heuristic across depths 4, 6, and 8[cite: 2].

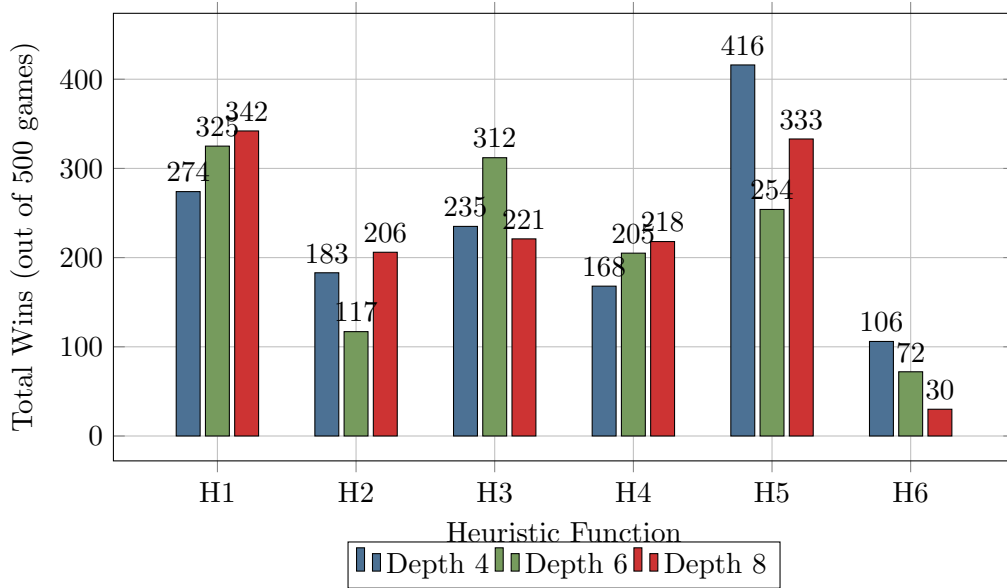


Figure 3: Total win distribution across search depths 4, 6, and 8.

5.2 Depth Progression and Scaling Trends

Figure 4 illustrates how performance scales as search lookahead deepens[cite: 2].

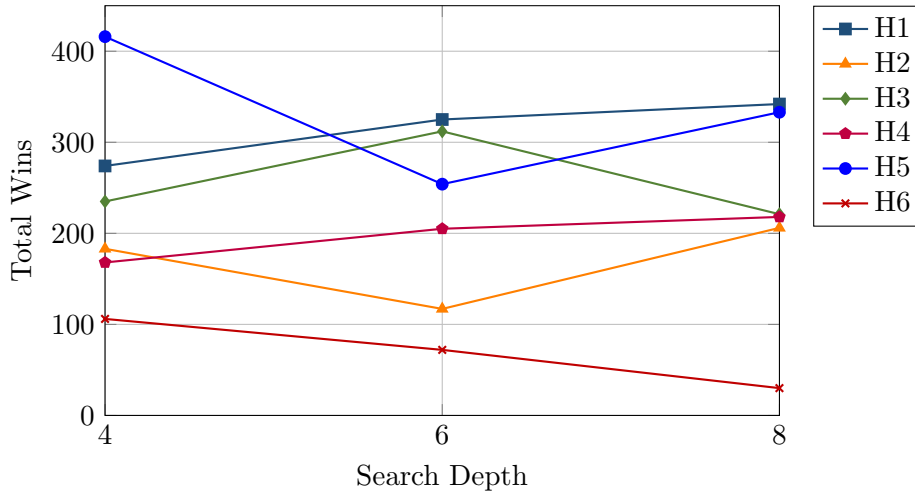


Figure 4: Win rate trajectory across lookahead depths showing steady scaling in H1 versus degradation in H6.

5.3 Draw Rate Analysis

Figure 5 highlights aggregate draw occurrences, demonstrating increased draw frequency at deeper lookahead depths due to symmetric defensive play[cite: 2].

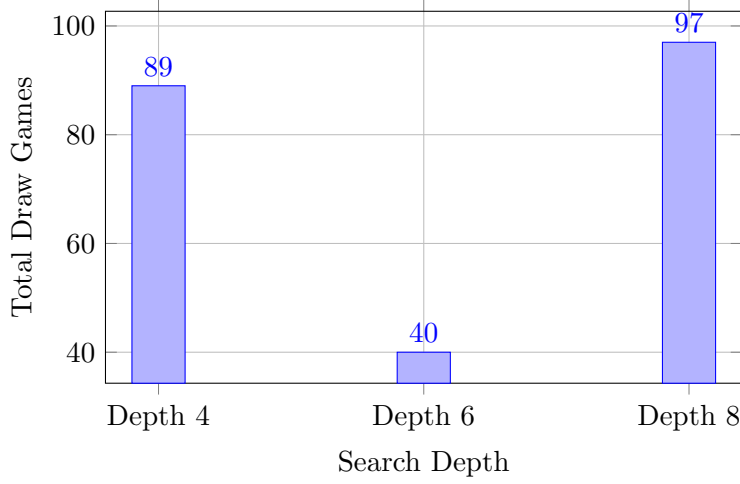


Figure 5: Total draws across all matchups per search depth.

6 Discussion & In-Depth Analysis

6.1 The Depth-Scaling Efficiency of Heuristic 1

A key outcome from the empirical dataset is the linear scaling performance of Heuristic 1 (H_1)[cite: 2]. Relying strictly on storage difference[cite: 1]:

- At Depth 4: H_1 accrued 274 wins[cite: 2].
- At Depth 6: H_1 accrued 325 wins[cite: 2].
- At Depth 8: H_1 accrued 342 wins[cite: 2].

Deep search tree traversal effectively compensates for simple evaluation formulas, allowing pure store difference metrics to outperform over-parameterized heuristics.

6.2 Dominance of Storage Proximity (H_5)

Heuristic 5 (H_5) performed exceptionally well at shallow depths, securing 416 wins at Depth 4 (83.2% win rate)[cite: 2]. Accounting for ‘stonesCloseToStorage’ allows the agent to position stones in pits 4 and 5, setting up extra turns and captures without relying on deeper search steps[cite: 1].

6.3 Performance Degradation in Non-Linear Models (H_6)

Heuristic 6 (H_6) dropped from 106 wins at Depth 4 to 30 wins at Depth 8[cite: 2]. The non-linear ratio $\frac{1}{24-s}$ generates steep evaluation spikes near win conditions, distorting intermediate node evaluation and impairing pruning efficiency.

7 Conclusion

The study demonstrates key trade-offs in heuristic design:

1. Positional features (H_5) provide strong advantages at shallow depths where tree depth is constrained[cite: 1, 2].
2. Simple functions (H_1) scale reliably with deeper search trees, providing consistent evaluation without multi-parameter noise[cite: 1, 2].
3. Non-linear distance functions (H_6) degrade performance by introducing score distortions that reduce Minimax efficiency[cite: 1, 2].